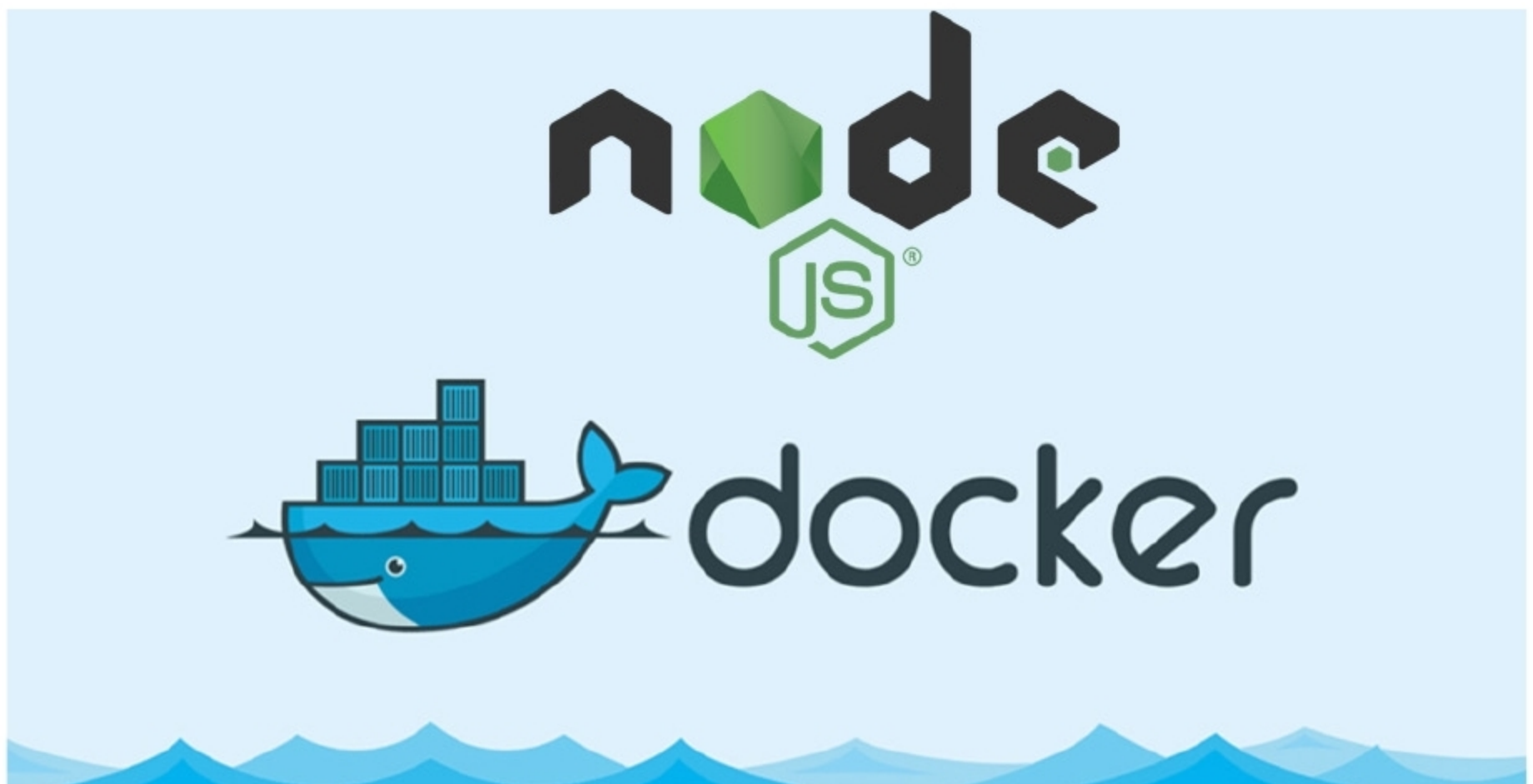


Dockerising a Node.js Application

Docker is a tool for packaging an application and its dependencies into a standardised unit, so that the app can be moved between different environments and run without any changes. This article details how a Node.js application can be Dockerised.



Containers allow developers to package an application with all of its dependencies, and ship it as a single entity.

Docker is a tool designed to ease the process of containerisation and is a part of many DevOps toolchains.

A Docker image is a read-only template of the changes in a root file system and other parameters for use within the container runtime. Basically, images are frozen snapshots of the container that can be made to run or stopped, whereas containers are the running instances of the images. A container is the isolated process in user space, which shares the kernel with other containers. When a container is started, Docker adds a read-write layer on top.

Node.js is an asynchronous event driven runtime that enables the development of scalable network applications. It uses an event driven, non-blocking I/O model that makes it lightweight and efficient for developing data intensive applications.

The need for Dockerising

We might have heard the following statement: ‘It works on my machine’ – it is from here that the need for Dockerising the application has come into the picture, for developers. Any Dockerised image build will run the same way it runs anywhere else. Using orchestration services, Dockerised applications can be used to respond to dynamic traffic demands as they are easily reproducible.

Setting up the environment

To set up the development environment, we will first need to install Node.js and Docker.

On Windows: The Node.js installer can be downloaded from <https://nodejs.org> and Docker for Windows can be downloaded from <https://hub.docker.com/editions/community/docker-ce-desktop-windows>.

On Linux: To install Node.js on Linux, use the following command:

```
sudo apt install nodejs
```

Once installed, the download can be verified by using the following command:

```
node -v
```